SRI GSC — Venture Brief

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Recover more iodine from what is already being produced — without pumping more brine.

On the SRI GSC Venture Brief template, 4 September 2026. Figures follow our business plan v1.9 (29 Aug 2026) and the pricing revision of 1 Sep 2026. "Estimate" means our own calculation from public data; "not yet measured" means open. The two Japanese producers we work with are named Producer A and Producer B because those discussions are confidential.

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1. Problem & Approach

What our future customers struggle with today. Japan is the world's second-largest iodine producer (USGS), and roughly 80% of that output comes from natural-gas brine on one gas field in Chiba Prefecture (prefectural production and water-use records for FY2024; our own figure is 82%). Those producers face a structural limit on growth: brine pumping is capped by ground-subsidence agreements between producers and the prefecture — we hold the text of those agreements — so the single input that sets output is bounded by regulation rather than by demand or capital.

Strategic-resource context. Iodine is a relatively small commodity, but its supply chain is strategically sensitive on two structural counts. **Supply concentration:** production is geographically concentrated worldwide and, within Japan, in a single gas field, and the element is difficult to substitute in several important applications. **Limited supply elasticity:** additional brine pumping faces physical and regulatory constraints, so rising demand cannot simply be met by pumping more. **Supply resilience** is therefore the opening we target — recovering more iodine from resources already being produced, rather than extracting more. We do not assume iodine is a designated critical mineral, and we do not argue that a shortage is under way; our thesis is narrower. Where supply is concentrated and extraction is constrained, secondary recovery from existing production can strengthen resilience.

At the same time, iodine remains in the discharge at concentrations that existing processes do not currently recover economically: blow-out and ion exchange pay at raw-brine concentrations of tens of ppm and stop paying below that. We separate what is published, what a customer has told us, and what we have inferred:

Basis	Figure
Published plant operating data	35 ppm brine; 60,000 t/day processed; 1,400 kg/day of iodine produced
Disclosed to us by Producer B	its discharge carries about 10 ppm
PhaseShift working estimate, derived from the rows above	about two-thirds of the incoming iodine is recovered; the remainder leaves with the discharge at 7–12 ppm; **700–1,000 t/yr uncaptured nationwide**, on the order of **JPY 10 bn/yr** at market price (700–1,000 t × JPY 11,782/kg = JPY 8.2–11.8 bn). Assumptions: the published plant is representative; discharge sits in the 7–12 ppm band; value at market price, not at a price we would receive
Being verified	the national figure, with the Society of Iodine Science

Recovering iodine from that discharge is one potential way to increase supply without pumping more.

How our technology addresses it, in plain language. A polymer developed at Doshisha University binds iodine at single-digit ppm; warming the water slightly makes it separate into a small concentrated phase that carries the iodine with it, and a further temperature step releases the iodine so the polymer is reused. Because the trigger is temperature, the regeneration step is designed to avoid a consumed reducing agent — a claim about regeneration only, since feed-side pre-treatment may still be required. The equipment is a side-stream unit on the discharge line: 3.5–6.8% of the plant flow is diverted into it, treated and returned, and the main line never stops. Within that diverted stream, only the concentrated phase — about 1–3% of it — is heated. The producer owns the unit and keeps the iodine; we supply the polymer and charge per kilogram recovered.

Why now. The producers opened the door: two majors have agreed to supply real wastewater, one with a written list of adoption requirements. Demand-side pull is strengthening in two places — context rather than the basis of the plan, and carrying no revenue in our financials: Japanese policy names iodine a principal raw material for perovskite solar cells intended for domestic manufacture, and iodine has demonstrated potential as an alternative propellant to xenon. And the same recovery structure exists at much larger scale in US produced water, where recovery at higher concentrations is already commercial.

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2. Technology Edge

What is genuinely new. The key question is no longer whether the polymer can capture iodine in a laboratory. It is whether that capture can become economically recoverable iodine from real wastewater in continuous operation. **Capture is not recovery.** Selective capture at single-digit ppm has been demonstrated in laboratory batch tests, including a model brine wastewater; performance in real high-salinity wastewater remains to be validated. What is novel is the release mechanism: desorption driven by temperature rather than by a consumed chemical. UV-vis behaviour (absorption maximum shifting 351 → 370 nm) indicates complexation with polyiodide species; the mechanism is still being characterised and we do not assert it as settled. Incumbent adsorption regenerates with a reducing agent — a recurring cost and a waste stream. Ours regenerates on a temperature swing using low-grade heat, and only the concentrated phase (about 1–3% of the diverted side stream) is heated — an engineering estimate of about 10.5 kW per unit rather than a measured plant figure — and one producer has disclosed an on-site steam-drain stream at 80–90 °C. Whether waste heat alone suffices depends on heat-exchanger efficiency and is not yet demonstrated.

Results and IP behind the claims (laboratory, batch, as of August 2026).

System	Conventional polymer (PVP)	PhaseShift agent	Basis
Iodine in hexane	73.9%	91.4%	capture
Pure water, 7 ppm (two conditions)	—	96.8% / 97.2%	capture, UV-vis 351 nm, supernatant
Model brine wastewater, 7 ppm	—	supernatant below detection limit	capture
Reuse, second cycle	—	79.4%	dose not recorded; cause unresolved

These are **capture** figures from the fall in supernatant absorbance. They exclude desorption and collection, so **recovery on real wastewater has not yet been measured**; our plan carries a 92.9% placeholder that we flag wherever it drives a number. **Research base.** The work originates in a Japanese university laboratory, as the call requires. The PI is an associate professor at Doshisha University, inventor of the foundational patent and a recipient of a joint-research award in materials and devices; a list of the laboratory's peer-reviewed publications underlying this chemistry can be supplied as supporting material. A foundational patent covering the polymer family is filed and published in Japan (Doshisha University, sole applicant; inventor: the PI). A first application patent, jointly filed by the university and Kato, cleared the university invention committee on 7 August 2026 and is scheduled for filing in October 2026; its content is withheld until then.

IP created during this programme. The call requires that IP newly created through its support vest, in principle, in a domestic Japanese entity. Being pre-incorporation, we will document before the sprint: laboratory results vest in Doshisha University under the exclusive-option arrangement we already use for public R&D; commercial outputs — market and customer data, commercial design — vest in the Japanese company on incorporation, held by Kato under an assignment undertaking until then. Nothing is structured to place new IP outside Japan.

Known gaps, and the engineering answer to each. These are the work, not footnotes.

1. **Polymer containment and discharge compliance.** The current liquid-liquid separation loses about 30% of the polymer per cycle (up to 50%) into the dilute phase — a consumables cost and, more seriously, a substance that must not leave with the discharge; our design limit is under 0.1 wt% carryover. Our proposed mitigation, not a solved problem, is a solid-precipitating copolymer — synthesised, loss rate not yet measured — plus a polishing trap on the return line.
2. **Continuous flow.** There is no continuous-flow data at all: the protocol uses static settling and centrifugation, while the plant runs continuously and Producer B has named continuous-flow compatibility its heaviest adoption requirement. A continuous contactor and phase separator, built as a test skid with an engineering partner we are recruiting, is our first hardware milestone.
3. **Speciation and salinity.** Tail water may carry residual sulfite, leaving iodine as iodide rather than the species we capture, so a controlled pre-oxidation step using the plant's existing infrastructure may be required; the requirement is being defined against real samples. Behaviour above 100,000 mg/L TDS is untested.

Differentiation. Blow-out and ion exchange work above our band, and neither they nor we have published performance data at 7–12 ppm, so we do not claim the band is empty. The closest patent, WO2025258648A1, targets 10 ppm and above and desorbs with a reducing agent; reverse osmosis loses rejection below about 15 ppm; Iofina's IOsorb resin operates on far richer water. The real alternative is that the customer keeps discharging. The economics of that differ by market, and we state both: in Japan the producer pays no disposal charge on this stream, so we remove no cost and must create new revenue; in the US the operator already pays to inject the water but receives nothing for the iodine in it. Either way the case has to rest on positive economics, not on compliance.

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3. Global Market Hypothesis

Why the US is the right test case. Not because the market is large, but because it tests whether the same recovery logic holds under different water chemistry, production structure and infrastructure — produced water from oil and gas rather than iodine-plant brine. If it holds, this strengthens iodine supply resilience across regions rather than being a Japan-specific wastewater technology. Whether the target water exists there at all is unproven, and is the first of the two questions below.

First market and customer type. Produced water in Oklahoma's Anadarko Basin. Two customer groups: water-midstream and saltwater-disposal operators whose aggregated produced water is not connected to an iodine plant, and the low-concentration streams downstream of plants that are. Iofina runs eight plants there (each 10,000–50,000 bbl/day) and has announced a Permian plant for late 2026 — evidence that iodine recovery from produced water is commercially viable **at higher concentrations**, not that our lower-concentration case works. From its disclosed output we estimate, as a PhaseShift calculation rather than a published figure, 59–108 ppm feed water leaving residuals comparable to or richer than Japan's tails.

Who actually decides, and what we have not tested. At a producer the case is assembled by a plant or technology manager who must satisfy three people: operations, whose first question is whether the main line ever stops; the environmental manager, who asks what leaves with the discharge; and finance, weighing payback against a capital budget set a year ahead. That is why the unit is a side stream, why polymer carryover is treated as a discharge question rather than only a cost, and why we quote payback. What we have **not** tested is willingness to pay: no producer has committed to buy a unit, and the paid field unit is that test. Infrastructure operators buy on someone else's operating data, so the order in which sites are won matters more than the size of the market.

Who we compete against, including doing nothing. Iofina — a potential competitor if it chooses to miniaturise and move down in concentration, and equally a potential partner, since our unit works on water its plants have already stripped. Incumbent Japanese processes, in our home market. The closest patent family. And above all disposal: today the operator simply injects the water, at USD 0.60–1.25/bbl. Iodine value equals disposal cost only at **51–106 ppm** (at USD 74.27/kg), which is why a dedicated plant is hard to justify below that band. Our leading commercial hypothesis follows from that arithmetic: a low-CAPEX retrofit owned by the water operator. It is a hypothesis, and Milestone 2 is designed to test it rather than to assume it.

Why we believe it, and what would change our mind. The economics of the band are arithmetic rather than opinion; one Japanese design site (about 13,000 bbl/day) gives a comparable flow-rate reference for initial US skid sizing, though process compatibility still requires validation; and the water is already aggregated — one operator alone reports 60 disposal facilities and 1.2 million bbl/day of capacity.

Two questions decide Phase 2, and they are why we are applying. 1. Does the target water exist at meaningful volume? Our central assumption is 20–40 ppm streams at 5,000+ bbl/day at aggregation points, and it is unproven. Oklahoma's commercial producing formation has been reported at 300–350 mg/L, which tells us richer water exists but not that thin water is collected anywhere. If it is not, the answer is No-Go, not "wait". **2. Does the polymer survive above 100,000 mg/L TDS?** Untested. If it fails, US produced water is closed to us.

We do not rely on critical-mineral subsidies or policy support for the US case.

Scale of impact we are seeking. In Japan our addressable customer-value pool is JPY 4.0–5.8 bn/yr at market price (49% of the national figure above: Chiba holds about 82% of output, the top five municipalities 59.8% of that), and one site recovers 4,316 kg/yr, adding JPY 50.85M of sales for its owner. Modest in revenue, meaningful in kind: additional supply of a resource Japan already leads, obtained without drilling more. The North American opportunity is larger in volume, though we have not sized it in a way we would defend, and we will not claim it before the two questions above are answered.

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4. Your Team

How the team formed. The chemistry is Associate Professor Shinnosuke Nishimura's at Doshisha University; he is the inventor on the foundational patent. Kato was introduced through Japan's university-startup executive-matching route in Kyoto and has worked with the laboratory since spring 2026, taking the commercial side: customer development, public-funding applications, and IP and contract strategy with the university. On 19 August 2026 Nishimura agreed to serve as principal researcher for our JST D-Global application; one research staff member runs synthesis and process validation. The division is deliberate — the PI remains a university researcher and technical advisor rather than an operating officer, and the commercial risk sits with Kato.

Current gaps and how we intend to close them. We state these plainly because three of the five are exactly what we hope to work on during the sprint.

Gap	How it gets closed
Technical leadership	PI plus one research staff member today; recruiting a technical co-lead. Active search
Engineering / manufacturing of the unit	No partner yet, and continuous-flow hardware is our first technical task. Partner search active; introductions welcome
US market access	No presence and no operator relationship. **A primary objective of this sprint**

Offtake for concentrated iodine in the US	Unknown. To be tested during the sprint (Milestone 3)
Corporate form	Pre-incorporation, with incorporation planned around our public-funding and technical-validation milestones rather than a fixed date

Commitment during the five-month sprint. The two roles are complementary by design, and neither is a side project.

- **Business lead (Kato) — 100%.** PhaseShift is his full-time work. He leads the sprint, owns overseas customer discovery, and attends the entire two-week Silicon Valley bootcamp in person.
- **Principal researcher (Nishimura) — about one day a week, the maximum a serving academic can commit under university rules, all of it directed at sprint milestones**: mentoring sessions, the experimental plan and analysis. Laboratory execution is staffed by a research associate, so the experimental loop continues between sessions rather than stopping when he is teaching.

We would rather state a number we can hold to for five months than one that looks better on an application.

Compliance. We will support the checks on export control, research integrity and conflict of interest — current practice rather than intention: the business lead prepared export-control filings for an international joint-research application this August, and the university's research-integrity and IP procedures govern the laboratory side.

How we decide, and how we handle disagreement. Technical judgements are the PI's, commercial ones Kato's; what matters more is that we set numerical thresholds before we have the data and hold ourselves to them. We keep two things separate. The six milestones in the next section are what we test **during** the sprint. Separately, a set of pre-agreed conditions must hold **before** we would start a US feasibility study at all — domestic technical validation, a paid field unit or equivalent commitment, the patent filing, ring-fenced funding, dedicated staffing, a US partner candidate, a site candidate, cost-share and a third-party verification lab. **None of those conditions is met today**, which is why we are not yet pursuing US commercialization. When we disagree, the question becomes which measurement settles it and when we will have it. We keep a single register of every number and its source, so a figure cannot quietly diverge between documents; twice this year that discipline forced us to disclose things we would rather not — the 30% polymer loss, and the fact that our recovery rate is a placeholder.

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5. Milestones — what we will validate between October 2026 and February 2027

Each hypothesis carries both halves the call asks for: a **success criterion** — the observation that lets us proceed — and its **falsifier**, stated here because it is the harder half to write honestly.

#	Assumption to test	Success criterion / falsifier: how we will know it is wrong	Where in the program we test it
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A. Commercial feasibility			
1	Thin, unconnected produced-water streams exist at aggregation points in Oklahoma at volumes worth a unit	A concentration-and-volume map built from operator and state data shows no 20–40 ppm streams at 5,000+ bbl/day, or every such stream is already contracted to an existing plant	US customer discovery; mentor introductions; Silicon Valley bootcamp
2	A US water operator will own the unit and pay a per-kilogram fee	Operators tell us they will only accept a tolling or build-own-operate model, or will not take title to a chemical product	Customer interviews during the sprint and the bootcamp
3	Concentrated iodine has a buyer in the US	No refiner will state receiving conditions (form, purity, minimum lot) or an indicative price	Business-model mentoring; partner introductions
B. Technical feasibility			
4	The polymer functions above 100,000 mg/L TDS	Phase separation or selectivity fails on a high-salinity sample at a third-party lab	Third-party validation partner introduced through the program
C. Defensibility and financing			
5	Our IP position supports a US entry	Freedom-to-operate review finds our route blocked by IOsorb-type or competing adsorbent claims	IP strategy sessions, aligned with our October 2026 filing

6	The venture is fundable on a Japan-first, resource-recovery narrative	Deep-tech and climate investors tell us the milestones we plan do not de-risk what they price	Investor readiness sessions

Two results feed the sprint: a TRL4 assessment on real wastewater from Producers A and B in late September 2026 under a predefined go / conditional-go / pivot / stop rule, and our first patent filing in October 2026. **What we want to hand back in February is a defensible GO / PIVOT / NO-GO decision on North America, not a slide that says the market is large.** We are not asking SRI to validate a market-size estimate; we are asking for help resolving the technical and commercial uncertainties that decide whether North America is worth pursuing. We note that the programme may discontinue support against agreed criteria, and we welcome that: we already run the venture on pre-committed numerical gates, and a programme willing to stop funding a hypothesis it has disproved is applying the discipline we apply to ourselves.

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6. 3–5 Year Vision

Where we want to be by 2030–2031. The default retrofit for iodine in streams existing plants cannot economically treat, first in Japan and then in North America — in one line, a standard retrofit layer for iodine supply resilience. In Japan that is on the order of five installed sites in our base case, each recovering about 4.3 t/yr for its owner, with our revenue from unit sales, polymer supply at JPY 10M per unit per year and a JPY 2,000/kg-I_2 process fee: roughly JPY 293M in FY2030 on current prices. **This is an illustrative base case, not a forecast**, and we do not quote operating profit because the cost base is still being rebuilt. In North America, one field demonstration with a water-midstream operator.

What success looks like at scale. Installing becomes routine because the payback is short and the plant never stops: the design case gives the owner JPY 31.62M a year against a JPY 100M unit, a 3.16-year payback, about 2.2 years where Japanese investment tax incentives apply. Those figures rest on an unverified recovery rate, so here is the sensitivity rather than the headline: at 70% recovery payback is about 4.7 years, at 50% about 8.3 years — which no infrastructure operator would accept. That is why the September measurement is a gate, not a milestone. The recurring layers, not the hardware, become the business. **Strategic importance is not willingness to pay.** No operator buys because iodine matters to the state: the purchase must clear recoverable iodine value against unit cost, polymer and process fees, and we assume no price premium and no subsidy in any figure here. Supply resilience is what makes the outcome matter to policy and investors; it is not what makes a producer sign. To be precise about status: customer ownership of the unit is our current commercial hypothesis — the producer owns it, keeps the recovered iodine, and pays us for polymer and per kilogram recovered — and Milestone 2 exists to test whether operators will accept it rather than demand a build-own-operate structure. Beyond iodine, the same machine with a different binding group extends naturally — bromine from the same brine is the obvious first candidate — which we treat as an option and keep out of the financials.

Two-step trajectory: primary supply first, then circularity. Step 1 is this brief: the standard retrofit for iodine that primary production leaves behind, in brine and produced water, expanding supply without drilling more. Step 2 applies the same temperature-responsive separation to iodine-bearing waste from perovskite solar-cell manufacturing and, later, end-of-life modules — same element, same chemistry, different feed — which Japan's intent to manufacture those cells domestically makes a plausible second market. **Step 2 is a concept: unproven, no experiments run on perovskite process waste, and no revenue from it in any of our financials.**

What has to be true. Recovery verified on real wastewater rather than assumed; continuous-flow hardware that runs unattended; polymer loss low enough that consumables are a margin, not a leak; producers who buy rather than wait; and, in the US, thin streams that are real and a polymer that survives the salt. We would rather reach February 2027 having disproved one of these than having avoided testing it.

What we can contribute to the programme. The call expects selected teams to be partners in building a Japanese model for deep-tech commercialization, not only recipients. Two things we can put in: a documented gate framework that decides go/no-go on numbers fixed in advance, which we already operate and would open to the programme; and a deliberately unglamorous case — a materials venture selling into conservative infrastructure operators, where the hard part is adoption rather than invention. If the model works only for software-shaped ventures it will not travel across Japanese deep tech, and we would rather be the awkward test case than the easy one.

The sequence we are asking to be judged on is technical validation, then commercial validation, then — only if both hold — additional domestic supply obtained without additional extraction. **SRI is therefore not being asked to validate a market-size story. We are asking it to help us kill or validate the assumptions that decide whether this can become a globally deployable resource-recovery business. By February 2027, we intend to know whether that proposition survives real wastewater, industrial salinity, continuous operation and customer economics.**

End of brief.